



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



3.2 Part 2: Supervised Learning Algorithms

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

Lesson Plan

- 3.2a: Introduction to Machine Learning
- 3.2b: Supervised Learning Algorithms
 - Linear Regression
 - Logistic Regression
 - K-Nearest Neighbours (KNN)

Introduction to Linear Regression

A statistical/machine learning model for regression

Model relationship between dependent variable (Y) and independent variables (X)

Components:

Y : Dependent variable
X : Independent variable
 β : Coefficients
 ε : Error term



$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

Linear Regression Example

Predicting Salary (Y) based on:

X_1 : Years of experience

X_2 : Education level (in years)

$$\text{Salary} = \beta_0 + \beta_1(\text{Experience}) + \beta_2(\text{Education}) + \varepsilon$$

β_0 : Base salary with zero experience/education.

β_1 : How much salary increases per year of experience.

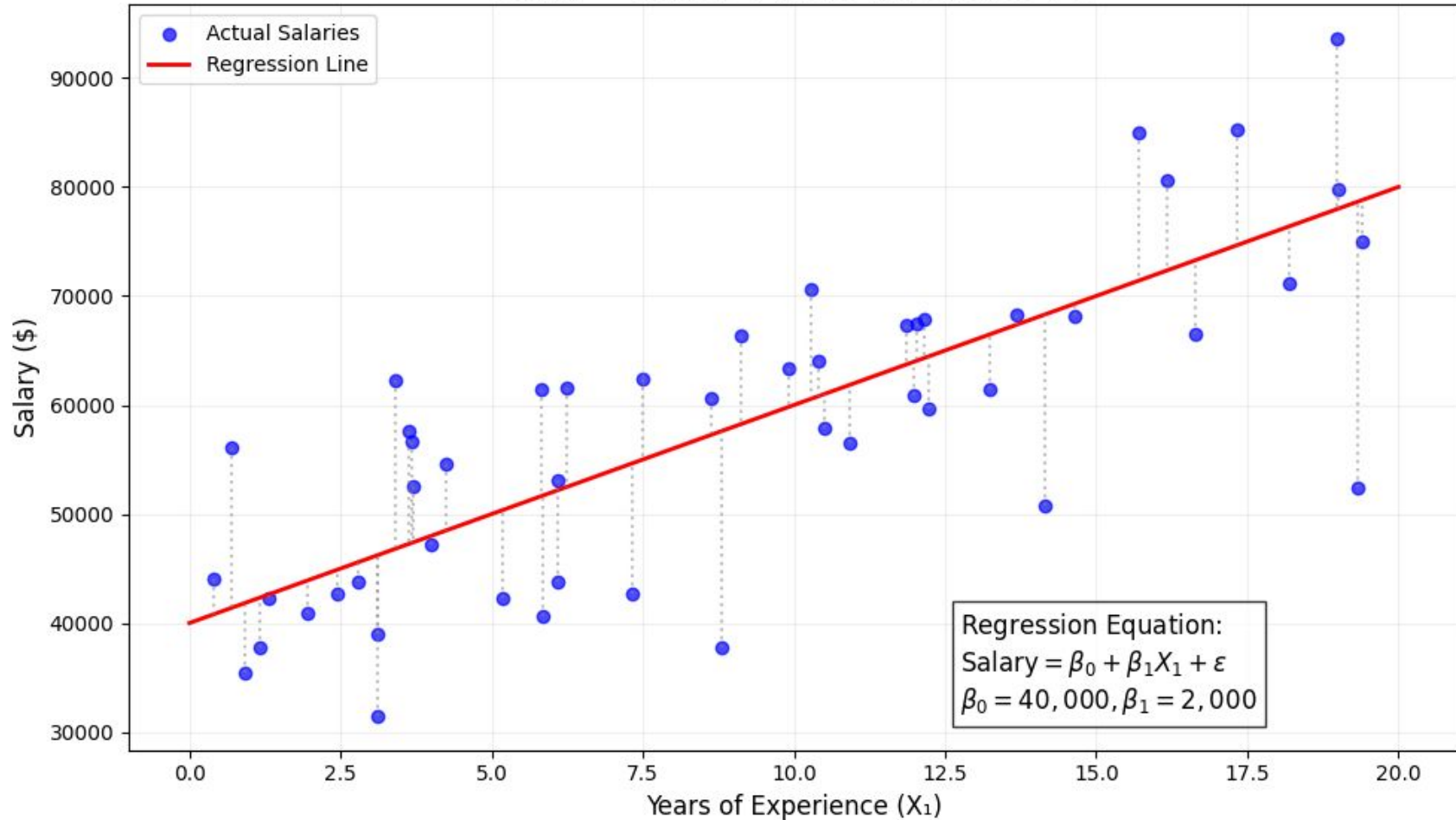
β_2 : How much salary increases per year of education.

ε : Unpredictable factors (e.g., negotiation skills, market demand).

Key assumptions:

- Linear relationship between variables
- No multicollinearity among independent variables
- Homoscedasticity: Constant variance of errors
- Normal distribution of error

$$\text{Salary} = 40,000 + 2,000 * \text{Experience} + \epsilon$$



Training the Model

Cost function: *Mean Squared Error (MSE)*

Find coefficient values that minimize squared differences

Methods:

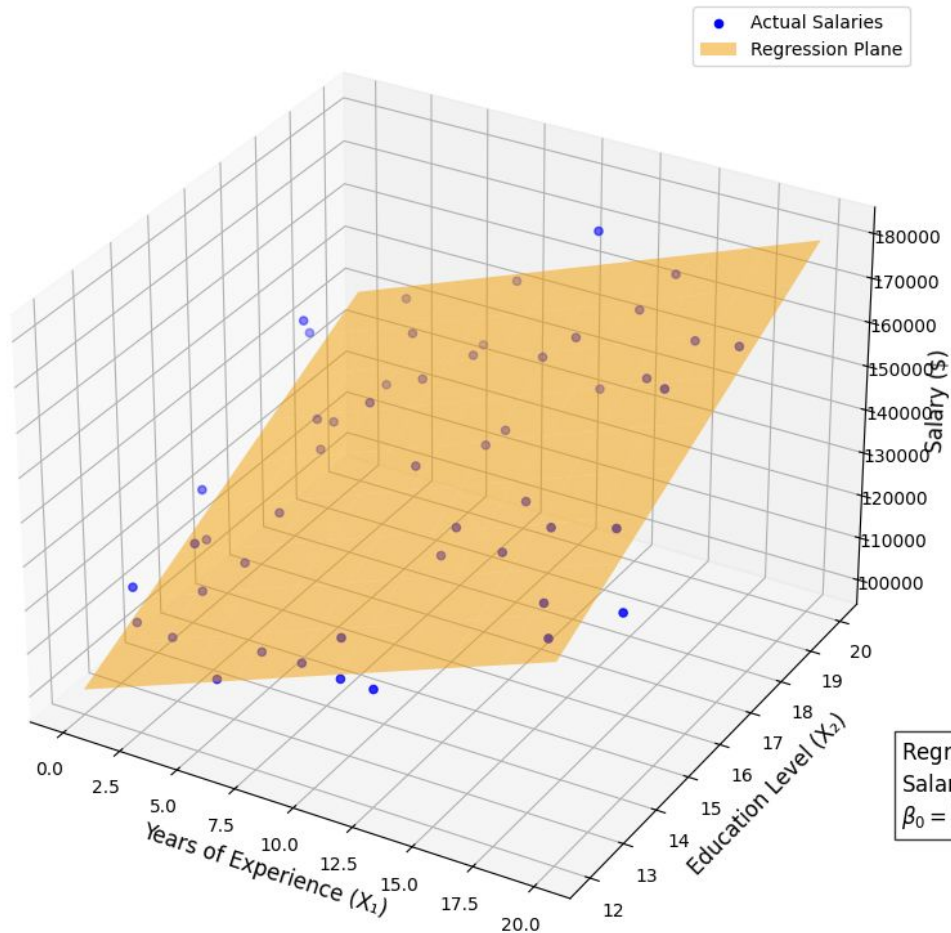
Ordinary Least Squares (OLS), Gradient Descent

Process:

Iteratively adjust coefficients to minimize error

$$MSE = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\text{Salary} = 40,000 + 2,000 \cdot X_1 + 5,000 \cdot X_2 + \varepsilon$$



Regression Equation:

$$\text{Salary} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

β_0 = Base Salary, β_1/β_2 = Coefficients, ε = Noise

Sum of Squares in Linear Regression

Sum of Squares Total (SST)

Measures total variation
in dependent variable

$$SST = \sum (y_i - \bar{y})^2$$

SST is your baseline for understanding how much “noise” exists in the data.

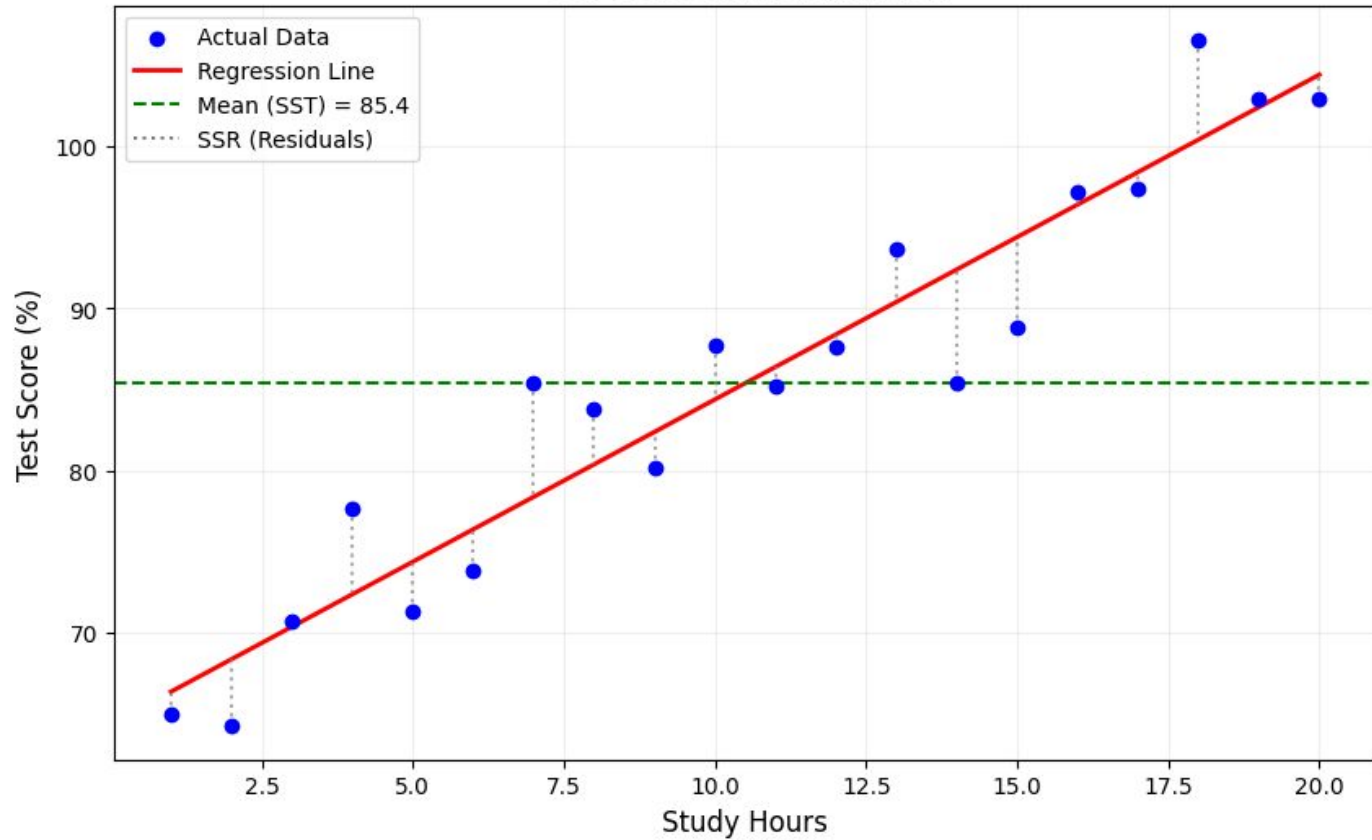
Sum of Squares of Residuals (SSR)

Measures unexplained
variation

$$SSR = \sum (y_i - \hat{y}_i)^2$$

SSR tells you how much variation your model **failed** to explain. A smaller SSR means your model fits the data better.

SST vs. SSR Visualization



R-Squared (R^2) in Linear Regression

SST = Total variation

SSR = Residual variation

$$R^2 = 1 - (SSR/SST)$$

Value range: 0 to 1

Interpretation:

Higher R^2 indicates better fit

Limitation:

High R^2 doesn't guarantee appropriate model
(overfitting risk)

Summary

- Key concepts recap:
 - Linear equations
 - Assumptions
 - Evaluation metrics
 - Common applications:
 - Demand forecasting
 - Financial analysis
 - Scientific research
- Extensions:
 - Polynomial regression
 - Regularization techniques

References

<https://mlu-explain.github.io/linear-regression/>

<https://mlu-explain.github.io/train-test-validation/>

<https://mlu-explain.github.io/logistic-regression/>

<https://www.luseratech.com/knn-visual>

End of Lesson - Exit Ticket

Survey Link

<https://www.menti.com>

